

Hand-in assignment 1

Instructions

- This assignment consists of **two questions**.
- On Vula it opens at 12:00 on Friday, 01 March 2024, please upload your solutions under **Vula Assignments no later than 23:55 on Tuesday, 05 March 2024**. Make sure that no pages or questions are missing.
- Write your full name and student number on your solutions.
- If possible, **upload a single document** saved as **ABCXYZ001_A1** (where **ABCXYZ001** is your campus Id).
- Show all your work or reasoning. **Answers without work or reasoning shown where needed will receive no mark.**

Questions

1. State whether the following are true or false. If true, prove it or if false, give a counter-example.
 - a) If (X, d) is both a quasi-metric space and a pseudo-metric space, then (X, d) is a metric space. [3]
 - b) Let (Q, d) be a quasi-metric space. If a sequence (x_n) right converges to x , that is, sequence $(d(x_n, x))$ converges to 0, then x is unique. [3]
 - c) In any metric space, arbitrary union of closed subsets is closed. [3]
 - d) In any finite metric space, every convergent sequence is eventually constant. (A finite metric space (X, d) is one in which the set X has only finitely many elements in it). [3]
 - e) In any metric space (X, d) , if A and B are subsets of X , then $\overline{A \cap B} = \overline{A} \cap \overline{B}$. [3]
2. Given a metric space (X, d) . Let

$$d^*(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Show that (X, d^*) is a metric space. (Hint: You may need to check and use the fact the function $f : [0, \infty) \rightarrow [0, \infty)$ defined by $f(t) = \frac{t}{1+t}$ is increasing). [6]